Ryan Zheng

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Education

University of Illinois Urbana-Champaign Urbana-Champaign, IL
Bachelor of Science in Computer Science | GPA: 3.94/4.0 August 2025 – May 2028

- Relevant Coursework: CS 225 Data Structures, CS 233 Computer Architecture
- Upcoming: CS 341 System Programming, CS 498 Robotics Team Project, ECE 598 Dynamical Sys. & Neural Networks
- Top 8 of ∼90 teams in CS 233's SpimBot tournament – competitive MIPS assembly optimization

Experience

Research Intern May 2026 – August 2026
Army Educational Outreach Program – NeuroAI Lab, Penn State

- Implemented a Multi-Agent PPO (MAPPO) training pipeline — shared-parameter actors, centralized critic, and a custom per-agent rollout buffer — inside a multi-threaded C++ inference server, extending it to route concurrent multi-agent requests via type-safe dispatch
- Designed an AR(1) autoregressive smoothed-noise domain-randomization scheme for central pattern generator (CPG) phase robustness, prototyped and validated in a standalone Python simulation before integrating into the C++ trainer
- Created a dedicated evaluation thread in a parallelized C++ RL trainer, enabling deterministic, periodic tracking of CPG phase and foot-contact dynamics without interrupting the main training loop.

Software Engineer February 2026 – May 2026
UIUC Disruption Lab – Urbana, IL

- Evaluated LLM inference serving frameworks (llama.cpp, vLLM) and benchmarked speculative decoding parameters on a distributed cluster of legacy iMacs with AMD Radeon Pro 580X GPUs, repurposing idle hardware for local LLM inference
- Resolved GPU backend incompatibilities on macOS by sourcing pre-compiled Vulkan binaries for llama.cpp, restoring GPU acceleration on unsupported legacy hardware and improving inference throughput by two orders of magnitude

Project Lead September 2025 – Present
Sig:Robotics – Urbana, IL

- Built a time-synced EMG and computer vision data pipeline, increasing sampling rate between acquisition hardware and host from 80 Hz (BLE) to 250 Hz by implementing UDP protocol and tuning data buffering
- Designed EMG data collection experiments and built end-to-end scripts for data cleaning and feature extraction, unifying EMG signals and video-derived ground-truth arm joint angles into a single CSV dataset

Research Intern June 2024 – May 2025
TENNLab – University of Tennessee, Knoxville

- Improved autonomous navigation efficiency by 35% in the F1Tenth simulator through reward tuning and applying evolutionary algorithms to optimize Spiking Neural Network controllers
- Researched evolutionary optimization of Liquid State Machines for real-time radio signal modulation classification under Professor Catherine Schuman

Projects

C++ Tensor Autograd Engine May 2026

- Developed a custom n-dimensional tensor autograd engine in C++ from scratch, supporting dynamic computation graphs, operator overloading, and reverse-mode backpropagation for MLPs and CNNs.

Diffusion Model from Scratch May 2026

- Implemented a classifier-guided Denoising Diffusion Probabilistic Model (DDPM) in **C++** with minimal libraries, capable of generating MNIST handwritten digits, building supporting Adam optimizer, MLP, and CNN from scratch

Skills

Languages: Python, C++, Java
Technologies: NumPy, Scikit-learn, LibTorch (C++), MuJoCo, llama.cpp, vLLM, FastAPI, Arduino, Git, Linux/Unix